Supplementary material

Data provenance, fixed experiments, numerical diagnostics and verification receipts

S1. Data eligibility and provenance

The official Figshare v2 MAT file is the processing source. The complete JSON was independently checksum-validated and 100 seeded raw coordinates were checked against the flattened MAT output. This verifies format consistency, not original specimen records. The original Centola 518-row dataset and exact code were not obtained.

Stage	Rows	Conditions	DOIs / blocks
All S-N data	15146	1610	459 DOIs
Ordinary316L L-PBF	1473	160	46 DOIs
Primary eligible stress control	1126	120	37 DOIs /29 blocks
Failures / runouts	987 /139		
Near-point removal sensitivity	942		29 blocks
Weak-edge grouping sensitivity	1126		28 blocks

Missing metadata in the preliminary eligible cohort

Scalar field	Missing rows	Interpretation
power	548	Training-only finite median + missing flag
scan_speed	533	Training-only finite median + missing flag
hatch_space	684	Training-only finite median + missing flag
layer_thickness	140	Training-only finite median + missing flag
direction	23	Training-only finite median + missing flag
heat_temperature	816	Training-only finite median + missing flag
heat_time	816	Training-only finite median + missing flag
load_ratio	8	Training-only finite median + missing flag

Missing counts above apply to the1126-row current primary cohort. All primary coordinates are approximate figure digitizations. Ambient temperature, uniaxial mode, Kt=1 and other entries can be database defaults. Absence of a scalar heat temperature can also mean no heat treatment; thermal categories remain separate.

Original-source validation covers six ordinary 316L points from Stern et al. and thirteen points from Voloskov et al., including four VHCF points outside the primary cohort. All19 agree in runout polarity; maximum coordinate discrepancy is approximately2.60% in cycles and0.25% in amplitude. The original raw experimental records were not obtained. Data are CC BY 4.0. Source DOI overlap is16of 17 Centola publications in ordinary 316L L-PBF and12 in the primary cohort; source overlap is not row identity.

S2. Fixed implementation and numerical behavior

Each main outer fold fits19 provenance blocks, calibrates on9 and tests on1. All87 main linear fits, 87 Basquin fits and 696 source-bootstrap fits converge. All87 primary1500-iteration MLP fits reach the cap; they are numerical diagnostics. A separately dated post-result extension at12000 iterations converges87/87, with maximum9183 iterations. All retain the original objective, inputs, regularization and seeds.

MLP analysis / bound	Violation (%)	Cost (decades)	Increasing-life slope (%)
Primary cap 1500 / model_q10	43.04-47.69	0.318	10.84
Primary cap 1500 / endpoint_pooled	10.52-10.98	1.824	10.84
Primary cap 1500 / cohort_max	0.52	4.118	10.84
Extended cap 12000 / model_q10	40.17-44.93	0.314	10.92
Extended cap 12000 / endpoint_pooled	9.30-10.06	1.886	10.92
Extended cap 12000 / cohort_max	0.23-0.25	4.214	10.92

Cost and risk definitions match Table1. Positive stress-location derivative violates the expected fatigue trend with other inputs fixed; the unconstrained MLP is not a validated design law. No comparison with the original variational BNN is implied.

Output magnitudes matter

Primary model / cohort max	Pooled median lower life (cycles)	Below one cycle (%)
basquin	917.5	0.00
linear	141.4	4.23
mlp	51.1	14.77

Pooled magnitude diagnostics repeat each test observation over3 seeds; they are not independent specimen counts. A nonvacuous but extremely small bound can still be unusable for high-cycle design. No post-result design-life target was optimized.

S3. Prespecified sensitivities: rich linear AFT

Analysis / bound	Specimen risk (%)	Cohort risk (%)	Cost (decades)
primary / cohort	0.20	4.60	3.54
primary / endpoint	14.50-15.40	52.87-55.17	1.13
primary / q10	28.97-30.77	71.26	0.62
deduplicated / cohort	0.22	4.60	3.56
deduplicated / endpoint	15.00-15.72	54.02-56.32	1.14
deduplicated / q10	28.91-30.71	71.26	0.62
weak edge / cohort	3.67-3.94	17.86	2.58
weak edge / endpoint	16.03-16.88	50.00-52.38	1.13
weak edge / q10	28.49-30.41	66.67	0.64
naive censoring / cohort	0.18	5.75	3.31
naive censoring / endpoint	15.57-16.22	54.02-55.17	1.01
naive censoring / q10	27.51-29.09	70.11	0.58
random rows / cohort	0.17	4.60	1.44
random rows / endpoint	11.38-12.31	41.38-42.53	0.65
random rows / q10	11.04-12.08	39.08-41.38	0.65
doi only common blocks / cohort	3.03-3.25	18.39	2.26
doi only common blocks / endpoint	20.09-21.47	65.52	0.63
doi only common blocks / q10	21.21-22.44	66.67	0.62

DOI-only predictions are reaggregated over the same29 primary blocks. Random-row tests withhold only20% of rows, so their test composition and sampling differ. Deduplication changes observations; weak-edge grouping changes group IDs, seeded fitting/calibration membership and block weighting. These differences are diagnostics, not isolated causal leakage effect estimates. Naive-censoring fits use incorrect training events but retain true censoring for evaluation.

S4. Gate contrasts and selected-source availability

Fraction / comparator	Known-risk difference (pp)	95% descriptive interval (pp)	True-risk identification (pp)
0.5 / distance	-0.307	[-2.097, +1.858]	[-0.481, +0.218]
0.5 / instability	-0.513	[-1.665, +0.519]	[-1.171, +0.012]
0.5 / random	-0.761	[-3.362, +1.552]	[-1.704, -0.236]
0.8 / distance	+0.130	[-0.639, +0.891]	[-0.348, +0.790]
0.8 / instability	-0.063	[-1.342, +0.909]	[-0.907, +0.597]
0.8 / random	-0.377	[-1.165, +0.273]	[-1.283, +0.284]

Negative favors the combined score. At the primary0.8target every descriptive interval crosses zero; not all seedwise signs favor combination. The primary strong claim fails. The0.5 random comparison has a negative empirical identification interval but a descriptive interval crossing zero; it cannot rescue the primary hypothesis.

Operational rule	Mean availability (%)	Blocks with any acceptance	Accepted risk (%)
combined_0.5	61.31	25/29	17.18-17.61
combined_0.8	85.77	28/29	15.02-15.82
distance_0.5	61.32	24/29	10.81-11.40
distance_0.8	82.25	28/29	13.92-14.69
instability_0.5	53.51	27/29	14.61-15.21
instability_0.8	85.53	28/29	13.61-14.33

Accepted risk is undefined for empty selections. It is conditional on blocks with usable observations, whereas availability includes zeros. Different accepted block/stress/runout mixtures prevent claiming calibration from these aggregate selected rates alone.

S5. Complete provenance-block results

Block (full mapping in source_family_map.csv)	N	Endpoint violation (%)
10.1002/adem.202200751	6	83.33
10.1007/978-3-030-31154-4_75	12	0.00
10.1007/s40964-021-00164-8	52	0.64
10.1016/j.engfracmech.2022.108883	10	0.00
10.1016/j.ijfatigue.2019.03.022	28	2.38
10.1016/j.ijfatigue.2019.105301	21	3.17
10.1016/j.ijfatigue.2020.106063	28	36.90
10.1016/j.ijfatigue.2021.106207	19	12.28
10.1016/j.ijfatigue.2021.106552	98	3.06-3.74
10.1016/j.ijfatigue.2022.106946	43	0.00
10.1016/j.ijfatigue.2022.107025	26	0.00
10.1016/j.ijplas.2021.103172	4	0.00
10.1016/j.jmst.2019.08.047	30	0.00
10.1016/j.jmst.2021.10.006	13	17.95
10.1016/j.msea.2020.140660	49	2.72-4.76
10.1016/j.mtla.2022.101591	8	79.17
10.1016/j.proeng.2018.02.002	24	20.83-34.72
10.1016/j.tafmec.2020.102849	8	0.00
10.1051/matecconf/201816502006	24	31.94-33.33
10.1080/00325899.2021.2010932	13	0.00
10.1108/13552541311302932	36	14.81-17.59
10.1111/ffe.13029	14	9.52
10.1111/ffe.13077	14	33.33-35.71
10.3390/ma12244203	56	19.64-20.83
10.3390/ma13153293	9	14.81
10.3390/ma15186278	18	0.00
curve_reuse_family_01	217	26.42
curve_reuse_family_02	199	5.53-5.86
curve_reuse_family_03	47	2.13-3.55

Means are across3 seeds, not independent repeated experimental cohorts. The three reuse families contain4,5and 2 DOIs; the remaining26 blocks are singleton publications. Physical independence and all possible reuse are not established.

S6. Verification, provenance and reproducibility

Eleven synthetic model tests check analytic likelihood gradients, Basquin recovery, censoring behavior, deterministic seeds, sample weighting and invalid inputs. Five evaluation tests cover finite-group rank resolution, strict censoring bounds, empty acceptance and training-only encoding. All16 passed before the experiment; they do not validate the physical model.

A separate implementation independently recalculates87 fold memberships, 30402 prediction rows and 2001 source/seed/policy cells. Training encoders and operational distance thresholds are checked for every fold; full location/bootstrap/calibration replay is sampled on the first source/seed0 and first multi-DOI family/seed42. All integrity checks pass. Numerical replay shares the estimator implementation and is not an external scientific replication.

The original first fold was rerun with exact agreement for54 predictions, including all3 models and 3 bounds. A complete end-to-end replay is also executed in results/full_reproduction; its comparison receipt accompanies the final package. The supplied run_all.sh creates a fresh timestamped run directory and rejects mismatched checkpoint signatures.

Protocol chronology

Version	Timing and purpose
v1	Pre-fit local protocol: outcomes, models, splits, calibration, gates and sensitivities
v1.1	Pre-fit minimum group arithmetic/checkpoint integrity/stress slope
v1.2	Pre-fit strict positive-claim criterion
v1.3	Pre-fit force-control eligibility and curve-reuse family blocks
v1.4	During execution, before aggregate inspection: explicit cohort target
v1.5	Post-result numerical failure investigation; fixed longer MLP cap

Local Git records support chronology but are not a public timestamp authority. Outcome-dependent protocol changes are explicitly labelled. Software, exact settings, seeds, splits, logs, raw hashes and CSVs accompany the package. No email, upload, public repository or publication was performed.

Primary claim receipts

Population: data/processed/grouping_report.json and primary.csv.

Main numbers: results/primary/summary.csv and source_seed_metrics.csv.

H2 failure: protocols/v1_2_prefit_claim_rule.md, results/primary/paired_contrasts.csv and independent_audit.json.

Sensitivity and convergence: results/sensitivity_comparison.csv; results/mlp_extended/fit_statuses.csv.

Raw and source evidence: data/raw/manifest.json; literature/primary_paper_audit.md; literature/followon_audit.md; literature/data_source_validation.md; literature/novelty_audit.md.